

Xiaoqing PAN

xiaoqing.pan@uni-jena.de | [LinkedIn](#) | [ORCID](#) | [Website](#) | +49 15563309561 | Jena, Germany

Department of Root Biology and Symbiosis, MPI of Molecular Plant Physiology

Re: Postdoctoral Position, SymbioticExchange Project

Dear Prof. Gutjahr,

My PhD was on RNAi and small RNA biology in a filamentous fungal pathogen. What draws me to your lab is that the questions your group is now asking about cross-kingdom RNA communication and extracellular vesicle-mediated signaling between plants and fungi are questions I have been working on from the other side of the interaction.

Your recent Nature Plants paper caught my attention because the experimental logic sits directly in my wheelhouse. The core of the work, pulling down plant AGO1 to identify which sRNAs are loaded during fungal colonization, and then asking whether some of those originate from the fungus itself, is the kind of question my PhD prepared me to think about. I was studying how *Aspergillus fumigatus* produces small RNAs, how its dicer-like proteins process dsRNA substrates, and how the resulting 21-nt 5'-U-containing sRNAs are loaded into argonaute complexes. Reading your AGO1 co-IP approach and the stem-loop RT-PCR validation strategy, I recognized the experimental thinking throughout. One open question your paper raises is whether the immune suppression that accompanies fungal colonization creates a broader vulnerability in the plant. If the symbiont is downregulating plant defense pathways to gain entry, does that open a route for opportunistic pathogens? My second PhD project addressed this kind of question from the host side: I built pipelines to analyze how human cells respond transcriptomically to three different fungal pathogens, characterizing pathogen-specific pattern recognition receptors (PRRs) signatures, alternative splicing, and immune gene dysregulation. That cross-system perspective on how hosts calibrate immunity during fungal interactions is not typical for plant biology candidates, and I think it connects to real open questions raised by your work.

The extracellular vesicle angle is where my neutrophil project becomes directly relevant. We analyzed RNA exported from primary human neutrophils during *A. fumigatus* confrontation, characterizing EV-associated sRNA, tRNA, and mRNA cargo from both the host and fungal fractions simultaneously. My hands-on experience analyzing EV-associated RNA cargo from both sides of a host-fungus interaction gives me a perspective I think would be genuinely useful in your lab.

My background spans both sides of what the SymbioticExchange project needs experimentally. On the wet-lab side, I have hands-on experience with sRNA cloning, custom sequencing library preparation, gene expression analysis by qPCR and Northern blot, genetic engineering of fungi, and GUS reporter assays from my master's work on rice-*Xanthomonas oryzae* interactions. My master's also gave me direct experience with plant cultivation, tissue preparation, and RNA extraction in a plant-pathogen system. On the computational side, I built pipelines on HPC for RNA-seq, sRNA-seq, and small RNA characterization, and I have experience analyzing sequencing data from both host and microbial fractions within the same experiment.

I want to be honest about the gaps. I have not worked with AM fungi directly, and transporter biology is new territory for me. The molecular mechanisms of phosphate and carbon exchange at the arbuscule interface are something I would need to learn from your team. What I bring is deep RNAi and small RNA expertise in a fungal system, plant molecular biology experience from my master's, hands-on EV-associated RNA analysis, and a host immunity perspective that connects to open questions your recent work has raised.

Best,

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SUMMARY

Doctoral candidate in Microbiology with deep expertise in fungal RNAi, small RNA biology, and host-pathogen interactions at the RNA level. Hands-on experience in sRNA cloning, custom NGS library preparation, RNA-seq and sRNA-seq pipeline development, and genetic engineering of filamentous fungi. Plant molecular biology background from MSc work on plant-bacterial pathogen interactions. Published five papers with two in preparation.

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Investigated the RNAi machinery of a filamentous fungal pathogen: characterized dicer-like protein specificity, argonaute loading, and small RNA biogenesis from inverted-repeat transgenes and endogenous loci
- Performed sRNA cloning, custom sRNA-seq and mRNA-seq library preparation, gene expression analysis by qPCR and Northern blot, and genetic engineering including generation of knockout and transgenic strains via Gibson assembly
- Coordinated a 20-person international collaboration across three institutions
- DOI: 10.1261/rna.079350.122 and DOI: 10.1101/2024.05.24.595671

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed how human immune cells respond transcriptomically to three fungal pathogens, characterizing pathogen-specific signatures in pattern recognition receptor pathways, alternative splicing, and A-to-I RNA editing
- Built HPC pipeline; wrote custom R scripts for multi-condition cohort analysis
- Led a 6-person cross-institutional collaboration with Fritz-Lipmann-Institut, Jena
- DOI: 10.1021/acsinfecdis.4c00598

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Performed sRNA-seq, tRNA-seq, and mRNA-seq analysis of RNA exported from primary human neutrophils during fungal challenge, characterizing both host and fungal RNA fractions simultaneously
- University of Jena collaboration

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Characterized sRNAs, lncRNAs, and circRNAs in rice responding to *Xanthomonas oryzae*; performed plant cultivation, tissue preparation, RNA extraction, and functional validation by qPCR and PCR
- Led a 5-person project with Sun Yat-Sen University, China; resulted in two published papers
- DOI: 10.1186/s42483-023-00188-8 and DOI: 10.1016/j.jplph.2022.153905

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango)

2026

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021
Thesis: Non-coding RNAs *ciR133* and *miR168a-5p* play roles in response to *Xoo* and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. *Mol Microbiol* 1-12.
 - **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. *ACS Infect Dis* 10(12), 4017-4029.
 - Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). *XOO*-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. *Phytopathol Res* 5, 22.
 - Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice *miR168a-5p* regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3*, *OsNPF2.4*, and *OsAGO1a*, respectively. *J. Plant Physiol* 208, 153905.
 - Kelani AA, Bruch A, Riveccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Knemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. *RNA* 29(7), 1033-1050.
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TECHNICAL SKILLS

- **Molecular Biology**: sRNA cloning, NGS library preparation, Gibson Assembly Cloning, Genetic Engineering of fungi, Northern Blot, qPCR, PCR, plant cultivation and tissue preparation, LC-MS, cultivation of microorganisms, GUS reporter assays
 - **Bioinformatics**: RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, Snakemake, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
 - **Programming**: R, Python, Bash, Git, Docker, Linux
 - **Language**: Chinese (native), English (fluent), German (B1)
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CONFERENCES

- **2025 - Oral presentation**: German speaking Mycological Society (DMyKG), Germany
 - **2025 - Poster**: Gordon Research Seminar - RNA Editing, Italy
 - **2024 - Oral presentation**: Jena RNA Club Symposium, Germany
 - **2024 - Poster**: RNA Society Meeting, UK
 - **2023 - Poster**: EMBL - The Non-coding Genome Symposium, Germany
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AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
 - Jena School for Microbial Communication (JSMC) travel grant 2023
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REFeree

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Knemeyer (olaf.knemeyer@leibniz-hki.de)